escape hatches are built in). More recently, several million-dollar heists in a DAO hack and other wallet issues were due to bugs in smart contract code. So smart contracts must be tested thoroughly before deployment for production use.

10.1.1 Types of testing

Software testing takes many forms, depending on the granularity of tests and time of test execution during the development phases:

- *Unit testing* --Testing of individpual components such as a single function
- *Integration testing* --Testing of the operation flow of the integrated system
- *System-level test-driven development* --Testing done to verify the integrity of the system developed by different members of the team as functions are added and checked into repositories

In previous chapters, for testing the Dapps, you used an ad hoc plan that exercised the operations of a Dapp by invoking them from the web UI. In this chapter, you focus on unit testing that involves exhaustive testing of the smart contract functions and modifiers. These tests are code scripts that simulate the execution of functions of the smart contract being tested. With the support of the Truffle test framework, the passing and failing of a test can be visually verified by check (✔) and X marks.